1. Target-blind Koblitz Frobenius-orbit factor-base input screen

2026-10-18. Exact finite input construction; no solver or end-to-end cost claim.

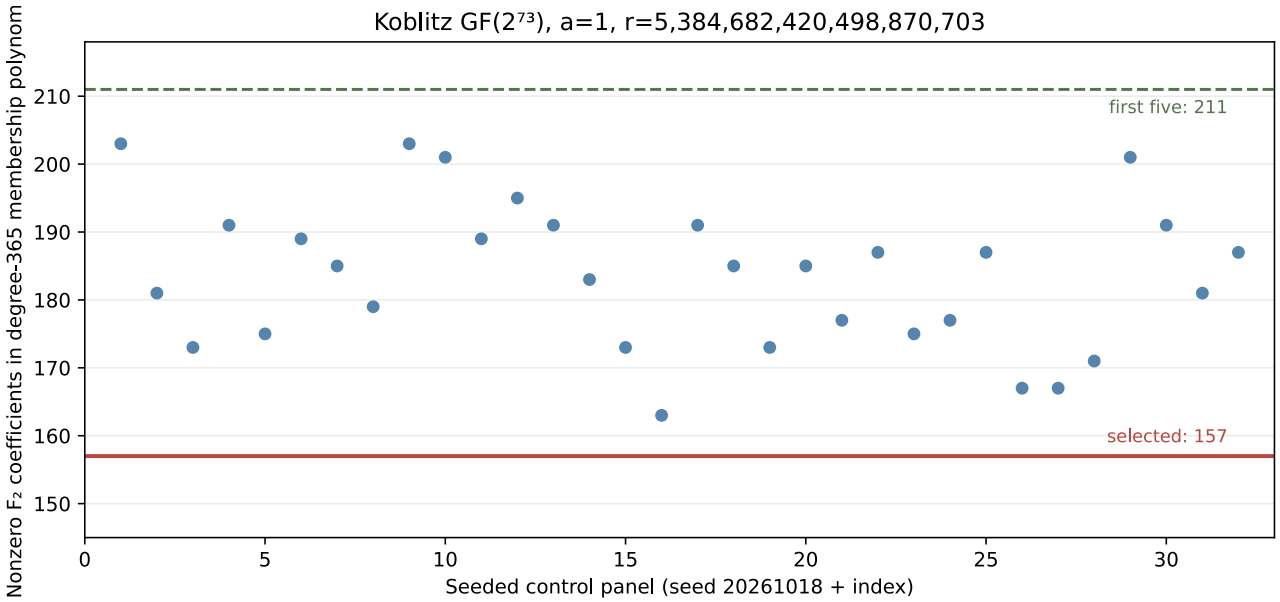
1.1. Fixed public input and method

The certified curve is $y^2 + xy = x^3 + x^2 + 1$ over `GF(2^73)`, represented by $z^{73} + z^{25} + 1$. Its target prime subgroup has order `r = 5,384,682,420,498,870,703` (`log2(r) ≈ 62.224`), with cofactor 1,754. The fixture and independent subgroup certificate are in `research/icperf_koblitz_subgroup_fixtures_20261017/`.

The producer examined raw x coordinates 1 through 512 before any target selection. Each liftable point P was projected to $F = [1754]P$. It kept one representative per distinct full Frobenius x orbit. For each 73-element orbit it computed the exact minimal polynomial over F_2. It greedily chose five orbits to minimize the nonzero coefficient count of their product, with ties resolved by raw x. The base has 365 materialized representatives and implicit negatives; its five raw seed coordinates are 498, 442, 41, 299, and 25.

1.2. Exact finite observations

Of 512 raw coordinates, 278 did not lift, none projected to the identity, and 14 duplicated an earlier projected x orbit, leaving 220 distinct candidates. The selected membership polynomial has degree 365 and 157 nonzero F_2 coefficients. Its supported exponents have maximum binary Hamming weight 7. The first five eligible orbits give 211 terms; 32 seeded five-orbit controls from the same pool give 163 to 203 terms, median 185. These are algebraic input counts, not solver timings or a performance claim.



The one-process construction wall was 39.628324 seconds, a descriptive setup observation. The exact bytes, selected points, controls, and source hash are in `research/icperf_koblitz_orbit_base_20261018/base.json`. Sage independently replays group identities, the finite candidate pool, polynomial coefficients, selection, and controls in `sage_replay.json`.

The 365 coordinates have at most 730 signed points. For an unordered eight-term sum with repetition, the exact multiset ceiling is $\binom{737}{8} = 2,078,085,779,887,029,220$, or 0.385925412 times r. The upper fraction across all arities one through eight is 0.390154731. Even a perfect solver on one unshifted five-orbit base cannot cover every subgroup target. Six full orbits have 438 coordinates and pass this necessary count threshold, but actual coverage remains unmeasured; shifts, extra bases and retries must be charged.

1.3. Group and phase binding

The subgroup Frobenius action is $\tau(P) = [\lambda]P$ with `lambda = 1,069,606,243,985,175,817 (mod r)`. The producer checked `lambda^2 - lambda + 2 = 0 (mod r)`, `lambda^73 = 1 (mod r)`, the generator identity, and all selected full-point identities. A later relation must retain its five unknown orbit logs and every relative phase. The raw points were chosen by x and cofactor projected; the construction supplies no discrete logarithms.

1.4. Limits and next check

The 157-term equation has degree 365. Its term count gives no measured degree of regularity, sparse solver cost, relation yield, useful rank, linear algebra, target work, or complete index-calculus time. The earlier complete Koblitz comparison uses a different curve and a 45-bit target subgroup, so it is not a timing baseline for this 62-bit subgroup. If a later solver uses raw-point predicates, it must include the cofactor and relevant torsion target channels.

A frozen experiment should retain this sparse five-orbit base and equal-size orbit controls, then construct a six-orbit arm before assessing target coverage. Compare chain-of-S_3 and direct S_8 seven-summand systems, and chain or S_9 eight-summand systems. Continue through natural relations, rank, target reconstruction, all charged costs, and matched rho before selecting an end-to-end winner. Timeouts remain censored.

Source and provenance: `tools/icperf_koblitz_orbit_base.py`, `research/icperf_koblitz_orbit_base_20261018/{base.json,sage_replay.py,sage_replay.json}`, and `docs/reports/2026-10-18-koblitz-orbit-base.md`. All are internal artifacts. The canonical ECDLP frontier graph remains unchanged because no reviewed attack-cost edge was established.